

FEATURE GEOMETRY FOR DATA-POOL SCREENING: UTILITY, STABILITY, AND LIMITS

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ABSTRACT

Selecting source-data pools for a target task requires deciding which combinations are worth training and validating. Frozen representations allow comparisons before supervised evaluation, but geometric diversity alone does not indicate whether the represented directions matter to the target. We study a two-stage approach that uses unlabeled features to form a shortlist and labeled validation to make the final choice. Starting from effective rank, constructed by exponentiating the entropy of normalized feature singular values, we examine how pooled features distribute variation across directions. We then compare target-weighted A-optimal design, motivated by a shared Bayesian linear model, with second-moment matching and target-blind diversity criteria. On PACS and Office-Home, direct selection produces small gains, and even the best evaluated combinations offer limited improvement over the baselines. In an equal-sample-cost DomainNet study, retaining roughly half the combinations preserves 98.3% of the test Brier top ten on average. Exploratory changes to projection seeds and the classifier readout preserve high average recall, including when candidate losses are more widely separated. Second-moment matching is competitive with A-optimal screening. These findings support using target-aware feature geometry to narrow the set of source combinations that receive supervised evaluation.

1 INTRODUCTION

Researchers adapting a predictor to a target domain often have several source-data pools to choose from. The pools may differ in acquisition conditions, image style, or the range of examples they contain. A source combination is useful only in relation to the intended prediction task, so its quality is usually assessed by training a model and evaluating it on target validation data. Repeating this procedure for every combination can be costly even when each individual model is small. The practical problem is deciding which combinations deserve that supervised evaluation.

Pretrained encoders provide a common space in which to inspect these choices. An encoder maps each image to a feature vector; with its parameters frozen, the same extracted features can be reused across source combinations and classifier fits. Learned visual representations such as CLIP and DINOv2 make this setting natural (Radford et al., 2021; Oquab et al., 2024). Stacking feature vectors as rows of a matrix exposes properties that can be measured without fitting a classifier. Sample counts indicate the amount of data, while singular values describe how strongly different feature directions are represented. These measurements offer a possible first step in a search that would otherwise require labels at every comparison.

Effective rank is one way to summarize that directional structure. Ordinary matrix rank counts nonzero singular values, so it can assign the same rank to a well-spread representation and one dominated by a few directions. Effective rank instead normalizes the singular values, takes their entropy, and exponentiates the result (Roy & Vetterli, 2007). It describes how many equally weighted directions would have the same spectral entropy. RankMe uses this idea to assess self-supervised representations without downstream labels (Garrido et al., 2023). For source-pool selection, the corresponding intuition is that a combination covering more directions may offer a richer set of features for learning.

Moving from representation assessment to pool selection introduces two questions. First, how do the directions supplied by different pools combine? Two pools can each have broad spectra while covering much the same part of feature space; another pair can complement one another. Second, which of the resulting directions matter to the target? A combination may have high effective rank because it contains variation rarely encountered at prediction time. For example, if the target occupies a narrow part of feature space, adding variation outside that part can raise a diversity score without improving its coverage. A pooled spectrum describes source variation, whereas prediction also depends on the target distribution and the relation between features and labels.

These questions suggest using geometry to retain alternatives rather than requiring it to choose the final model. A screen can keep several combinations whose features look relevant and let labeled validation resolve the remaining uncertainty. The resulting objective is to preserve useful candidates within a smaller evaluation budget. It differs from maximizing a geometric score for its own sake: a useful shortlist should contain candidates with good target outcomes, even when their exact ordering is hard to predict from features.

We study this approach through a two-stage data-pool screening protocol. The first stage ranks combinations using unlabeled source and target features; the second trains on source labels and chooses within the shortlist using separate target-validation data. Target-weighted A-optimal design favors source combinations that reduce uncertainty along directions common in the target, a connection made explicit by a shared Bayesian linear model. Second-moment matching asks a simpler question: how closely does the source’s average directional energy match the target’s? Comparing these scores with merged effective rank and other target-blind criteria distinguishes the use of target information from the choice of scoring function.

The experiments evaluate both the retained candidates and the final validation choice. Direct selection on PACS and Office-Home produces modest gains, and exhaustive evaluation finds little room to improve on the baselines within those candidate families. An equal-sample-cost Domain-Net study instead shows that target-aware screening can retain most promising combinations while shortening the list sent to supervised evaluation. Additional projection seeds and classifier readouts preserve high average recall, and second-moment matching remains competitive with A-optimal screening. By relating these observations to the construction of effective rank and the assumptions behind target-weighted risk, the paper identifies what geometry contributes to pool screening and what must be decided using task-specific evidence.

2 RELATED WORK

2.1 REPRESENTATION GEOMETRY AND DIVERSITY

RankMe uses effective rank to assess learned representations without downstream labels (Garrido et al., 2023). Determinantal point processes favor subsets with complementary directions (Kulesza & Taskar, 2012). Both motivate ways to compare source pools before fitting a task-specific model. Their geometric objectives differ, however, from preserving candidates with low target loss: variation may be redundant for a particular task, and directions important to that task may occupy only part of the representation. We examine merged effective rank and a specified subspace-DPP implementation as target-blind comparators.

2.2 OPTIMAL DESIGN AND DISTRIBUTION MATCHING

A-optimal design measures uncertainty through the trace of a covariance matrix, with applications to active linear regression (Fontaine et al., 2021). Weighting that covariance by a target feature moment directs the criterion toward predictions on the intended population. Distribution matching offers a different approach. MMD compares mean embeddings of source and target distributions (Gretton et al., 2012); a homogeneous quadratic kernel makes this comparison depend on their second moments. Moment alignment has also been used to learn representations for multi-source domain adaptation (Peng et al., 2019). Here the encoder stays fixed, allowing us to compare uncertainty reduction and moment matching as scoring rules on the same candidate combinations.

2.3 SELECTION UNDER A COMPUTATION BUDGET

Compute-constrained data selection accounts for the work needed to select data as well as the subsequent training (Yin & Rush, 2025). Pool screening introduces an intermediate decision: how many combinations should receive supervised evaluation? A broad shortlist may be easy to construct and preserve good candidates, while still requiring substantial validation work. Our protocol measures retention at a declared shortlist size and separates it from the quality of the final selected model. This provides the decision-level evidence needed before measuring the total cost of a deployment.

3 GEOMETRY-GUIDED SCREENING

3.1 PROBLEM AND TWO-STAGE PROCEDURE

Let $H_i \in \mathbb{R}^{n_i \times d}$ contain the features of source pool i , obtained with a fixed encoder and a shared, label-independent preprocessing map. Write $G_i = H_i^\top H_i$. A feasible combination S belongs to a finite family $\mathcal{C}$, with $G_S = \sum_{i \in S} G_i$ and $n_S = \sum_{i \in S} n_i$. An unlabeled target-selection sample estimates the uncentered second moment $C_T = \mathbb{E}_T[hh^\top]$.

The screen ranks $\mathcal{C}$ and retains its first s combinations as $\mathcal{Q}_s$. We then train predictors on their source labels and use a separate target-validation set to choose $\hat{S} \in \mathcal{Q}_s$. Target-test data evaluate this frozen choice and the candidates retained. The screening stage, denoted U0, sees source and target-selection features but no labels or validation/test data. The supervised stage, U2, reads source labels and target-validation data. Target-blind rules use only source information. This separation lets the screen exploit reusable features while reserving task-specific evidence for the final choice.

Here a pool is a fixed block of examples, and a candidate is a feasible union of such blocks. We select among these unions rather than choose individual images or retrain the encoder. This distinction fixes what a screening score is allowed to change. Each retained combination receives the same prescribed classifier and validation procedure, so differences in its outcome come from the source examples supplied to that procedure. In the equal-cost study, all combinations also contain the same number of examples.

3.2 CONSTRUCTING EFFECTIVE RANK FROM POOLED FEATURES

The singular value decomposition $H = U\Sigma V^\top$ separates feature directions from their strengths. The columns of V give directions in the encoder’s coordinate space, and the diagonal entries of Σ measure the magnitude of the feature matrix along those directions. Since $H^\top H = V\Sigma^\top \Sigma V^\top$, the Gram matrix contains the information needed to recover them. It also gives a convenient way to combine pools without repeatedly stacking their feature rows.

For fixed nonnegative weights w_i , vertically stacking source features gives

$$H_S = \begin{bmatrix} \sqrt{w_{i_1}} H_{i_1} \\ \vdots \\ \sqrt{w_{i_k}} H_{i_k} \end{bmatrix}, \quad H_S^\top H_S = \sum_{i \in S} w_i G_i. \quad (1)$$

The Gram sum therefore determines the merged singular values in the shared feature coordinates. Keeping these operators preserves both the strength of each direction and its alignment across pools. Re-centering or re-fitting a projection for each combination would change those coordinates and require a different calculation.

We construct the effective rank by normalizing the singular values into a probability vector and exponentiating its Shannon entropy (Roy & Vetterli, 2007):

$$r_{\text{eff}}(H) = \exp \left(- \sum_j p_j \log p_j \right), \quad p_j = \frac{\sigma_j(H)}{\sum_k \sigma_k(H)}, \quad (2)$$

The convention $0 \log 0 = 0$ handles zero singular values for a nonzero matrix. If ρ singular values are positive, the entropy lies between 0 and $\log \rho$, so $1 \leq r_{\text{eff}}(H) \leq \rho$. A rank-one matrix attains the

lower endpoint; equal positive singular values attain the upper endpoint. For two positive singular values with ratio 9 : 1, ordinary rank is two, but effective rank is about 1.38. Equal singular values instead give effective rank two. Thus the score distinguishes concentrated from balanced variation even when ordinary rank is unchanged.

In implementation, we first sum the source Grams, compute their nonnegative eigenvalues, and take square roots to obtain the singular values used in Equation (2). Normalizing the Gram eigenvalues directly would give a different score, with more weight on large directions. A common scalar rescaling cancels in the normalized singular values. In particular, repeating an entire feature matrix multiplies every singular value by the same factor and leaves effective rank unchanged. The statistic measures spectral shape, not the amount of labeled evidence available for training. Numerical tolerances are specified in Appendix B.

The direction information in the Gram sum is important before this spectral summary is taken. Keeping only each pool’s spectrum and a list of subspace overlaps can already lose information needed to calculate the merged effective rank.

Proposition 1 (Directional assignment matters). *There exist row-normalized pairs (A, B_1) and (A, B_2) with identical marginal singular spectra and identical principal angles between their row spaces, but different effective ranks after vertical concatenation.*

Proof. Take A with four rows $e_1^\top$ and one row $e_2^\top$. Let B_1 have four rows $e_1^\top$ and one $e_3^\top$, and let B_2 have one row $e_1^\top$ and four $e_3^\top$. All three nonzero singular spectra are $(2, 1)$; both row-space pairs have principal angles $(0, \pi/2)$. The merged Gram spectra are $(8, 1, 1)$ and $(5, 4, 1)$, giving effective ranks of approximately 2.6260 and 2.8495. $\square$

The two merges assign different amounts of energy to their shared direction. This explains why our screening rules use Gram operators rather than marginal spectra alone. Appendix A also gives an exact local spectral formula when the operators admit compatible two-direction blocks.

3.3 TARGET WEIGHTING AND MOMENT MATCHING

To connect geometry to prediction, consider scalar Bayesian linear regression with $\theta \sim N(0, P_0)$, $P_0 \succ 0$, and $y = h^\top \theta + \epsilon$, where independent noise satisfies $\epsilon \sim N(0, \sigma^2)$ and $\sigma^2 > 0$. Sources and target share θ and the noise model; target covariates reveal no further information about θ . Conditional on the source design, the posterior covariance and posterior-mean target risk are

$$P_S = (P_0^{-1} + \sigma^{-2} G_S)^{-1}, \quad R_T(S) = \sigma^2 + \text{tr}(C_T P_S). \quad (3)$$

The expectation averages posterior uncertainty and a new target observation. The derivation is recalled in Appendix A.

Equation (3) motivates the score $f_A(S) = \text{tr}(\hat{C}_T P_S)$, minimized with fixed hyperparameters. Source observations reduce uncertainty in the directions they cover, and the target moment weights those reductions by how often the directions occur at prediction time. Thus a pool can be valuable even without high overall diversity if it covers directions that remain uncertain and matter to the target. Adding a positive-semidefinite source Gram cannot increase this score.

For the experimental choice $P_0^{-1} = I$ and $\sigma^2 = 1$, write $G_S = U \text{diag}(\lambda_j) U^\top$, including zero eigenvalues. Then

$$f_A(S) = \sum_j \frac{u_j^\top \hat{C}_T u_j}{1 + \lambda_j}. \quad (4)$$

The denominator reflects source coverage of direction u_j , while the numerator measures its importance to the target. Two combinations with the same eigenvalues can therefore receive different scores if their directions align differently with the target. Effective rank treats their spectra identically; target weighting distinguishes where their coverage is useful.

Second-moment matching instead minimizes

$$f_M(S) = \left\| G_S / n_S - \hat{C}_T \right\|_F^2. \quad (5)$$

This is empirical squared MMD with kernel $k(h, h') = (h^\top h')^2$, including diagonal terms. It brings the source’s average directional energy closer to the target’s rather than weighting a posterior uncertainty matrix. Both criteria read the same features, so their comparison tests whether inverse-covariance weighting adds value beyond direct moment matching.

The shared conditional model supplies the risk interpretation of A-opt. Its assumption is substantive: if all features equal 1, source pools labeled constantly +1 and −1 produce predictors whose risk ordering reverses between two target worlds with opposite constant labels. The feature geometry is unchanged. In classification, we therefore evaluate the scores using trained ridge-softmax and logistic-regression readouts rather than identify Equation (3) with their losses.

3.4 SEPARATING RETENTION FROM FINAL SELECTION

Let $\ell(S)$ denote target risk for a fixed trained predictor and $v(S)$ its validation loss. If $|v(S) - \ell(S)| \leq \varepsilon$ throughout $\mathcal{Q}_s$, then the validation minimizer satisfies

$$\ell(\hat{S}) - \min_{S \in \mathcal{C}} \ell(S) \leq \underbrace{\min_{S \in \mathcal{Q}_s} \ell(S) - \min_{S \in \mathcal{C}} \ell(S)}_{\text{shortlist omission}} + 2\varepsilon. \quad (6)$$

The proof appears in Appendix A. The first term asks whether screening discarded the best available choice; the second concerns choosing within the retained set. A shortlist that preserves an oracle can still lead to an inferior final model if validation ranks its members differently. We audit their finite-test counterparts and also measure recall of the test top ten, which asks whether several good alternatives survive screening. The deviation condition is used to explain these two error sources, not estimated as a population guarantee in our experiments.

4 EVALUATION PROTOCOL

Controlled synthetic tasks test the risk model. PACS and Office-Home (Li et al., 2017; Venkateswara et al., 2017) then evaluate direct selection against a prespecified practical-effect threshold. An exhaustive headroom analysis follows those outcomes. The subsequent DomainNet study fixes sample cost and evaluates shortlists; its configuration was frozen before feature extraction and outcome evaluation. Test-sample resampling and the later projection/readout study are exploratory follow-ups. Appendix B records their configurations and result locations.

From the official cleaned DomainNet data (Peng et al., 2019), we select 100 class names by a fixed hash rule. Each of six domains contributes disjoint sets of 1,536 target-selection images, 1,024 target-validation images, and 3,072 anchor-pool images from official training data, plus 2,048 official test images. The manifest contains 46,080 images. Class metadata define the benchmark vocabulary but are hidden from screening.

CLIP ViT-B/32 (Radford et al., 2021) serves as a frozen construction anchor. Within each domain, its first principal-component scores define low, middle, and high strata, each supplying a block of 512 images. For a given target, we exclude its own blocks and enumerate $\binom{15}{3} = 455$ combinations, each with 1,536 source images. ResNet-50 and DINOv2-B/14 provide evaluation features (He et al., 2016; Oquab et al., 2024). All rules share row normalization, a fixed Rademacher projection to 128 dimensions, and post-projection row normalization. Holding sample count constant removes a direct source of training-cost differences between candidates.

Target A-opt uses $P_0^{-1} = I$ and $\sigma^2 = 1$. Comparators are second-moment MMD, target energy $\text{tr}(\hat{C}_T G_S / n_S)$, Bayesian D-opt, merged effective rank, subspace DPP, domain balance, and 20 seeded random rankings. They score the same candidate family. Domain balance additionally reads source-domain identifiers. Kernel definitions and tie rules appear in Appendix B.

The screen produces nested shortlists of sizes 10, 25, 50, 100, 227, with 227 designated primary. In the original protocol, U2 independently evaluates each method’s primary shortlist using ridge regression with regularization 1, no intercept, and a fixed softmax temperature. Validation Brier selects the final combination, breaking ties lexicographically. Smaller lists reuse that method’s nested evaluations; there is no cross-method cache. The test stage verifies upstream hashes before evaluating every combination to establish an exhaustive audit reference.

The follow-up adds four projection seeds and a multinomial logistic readout with fixed regularization. It keeps the candidate members and data roles unchanged. Screen, validation, and test remain separate processes, but the follow-up reuses each combination’s fitted model across methods to audit their shortlists. Its runtime is therefore the cost of a cached research audit rather than separate deployments. Readout settings and provenance checks are detailed in Appendix B.

We measure how many test top-ten candidates survive screening and how far the validation choice lies from the empirical optimum. Writing test Brier as $b(S)$ and its exhaustive minimizer as S^* , these quantities are

$$\text{Recall}_{10}(\mathcal{Q}_s) = \frac{|\mathcal{Q}_s \cap \text{Top}_{10}(b)|}{10}, \quad \text{Regret}(\hat{S}) = \frac{b(\hat{S}) - b(S^*)}{|b(S^*)|}. \quad (7)$$

The original acceptance criteria require at least 50% removal, mean recall of at least 90%, and mean normalized regret at most 5%. Brier is primary. Follow-up NLL and classification-error audits use their corresponding validation metric to select candidates, with the metric fixed before test access.

We average encoder results within each dataset–target-domain unit and random rankings within each task. PACS/Office-Home contain eight such units and DomainNet six. Follow-up seeds are also averaged within domains. Original bootstrap intervals resample target units; the test-sample analysis fixes trained models and resamples examples within each target. Seeds, encoders, and overlapping combinations are repeated measurements, not additional independent datasets.

5 EMPIRICAL FINDINGS

5.1 TARGET WEIGHTING UNDER THE SHARED RISK MODEL

The synthetic study varies feature dimension, candidate count, source budget, target sample size, and target rank, using 20 generation seeds for 216 design settings. Under the shared conditional model, A-opt reduces modeled risk by 4.23% relative to the strongest target-blind effective-rank comparator and by 14.85% relative to MMD. The respective descriptive 95% intervals are $[3.86, 4.62]\%$ and $[13.99, 15.70]\%$, resampling design settings after averaging seeds. These comparisons show the benefit of weighting uncertainty by the target when the model that motivates this operation holds.

A separate source-conditional-shift sweep changes the regression mechanism. Its mean normalized prediction regret increases from 0.211 without shift to 3.143 at the largest tested shift. This endpoint measures realized prediction rather than posterior risk. Together, the two studies distinguish the advantage of covering target-relevant directions from the effect of transferring labels across different conditional mechanisms.

5.2 DIRECT SELECTION AND AVAILABLE IMPROVEMENT HEADROOM

PACS and Office-Home test whether the score can directly choose a better source combination. At block budget $K = 3$, A-opt improves mean within-unit relative Brier by 0.351% over DPP and 0.273% over MMD. Both fall below the prespecified 2% practical-effect threshold. The paired difference interval crosses zero against DPP and is negative against MMD; detailed values are in Appendix C. These comparisons hold block count, rather than source sample count, fixed.

To understand the scale of the available gains, the post-hoc audit evaluates every feasible combination. Figure 2 shows that even the test oracle offers only 0.781% mean headroom over DPP. Most target units therefore leave little room for a different selector to improve the average loss substantially. This observation motivates examining candidate retention separately: geometry may help narrow the search even when the final choices have similar loss.

5.3 RETAINING PROMISING EQUAL-COST COMBINATIONS

Table 1 compares the original DomainNet shortlists. A-opt retains nearly all test top-ten candidates while discarding approximately half the combinations. Every primary shortlist contains a test Brier oracle; the remaining selected regret comes from validation choosing another member. This is the distinction captured by Equation (6): successful screening leaves a separate task for validation.

Table 1: The original DomainNet study retains $s = 227$ of 455 combinations. Means average encoders within six target domains. Recall concerns the test Brier top ten; regret is relative to oracle Brier. Spearman measures agreement between the complete screening and test-loss rankings.

Screen	Recall (%)	Regret (%)	Spearman
Target A-opt	98.333	0.001311	0.7443
Second-moment MMD	98.333	0.001311	0.6902
Bayesian D-opt	82.500	0.002658	0.4996
Merged effective rank	81.667	0.002909	0.4951
Random (20 repeats)	48.958	0.003899	0.0016
Target energy	46.667	0.009347	-0.0399
DPP subspace	27.500	0.009067	-0.0633
Domain balance	25.000	0.012083	0.0681

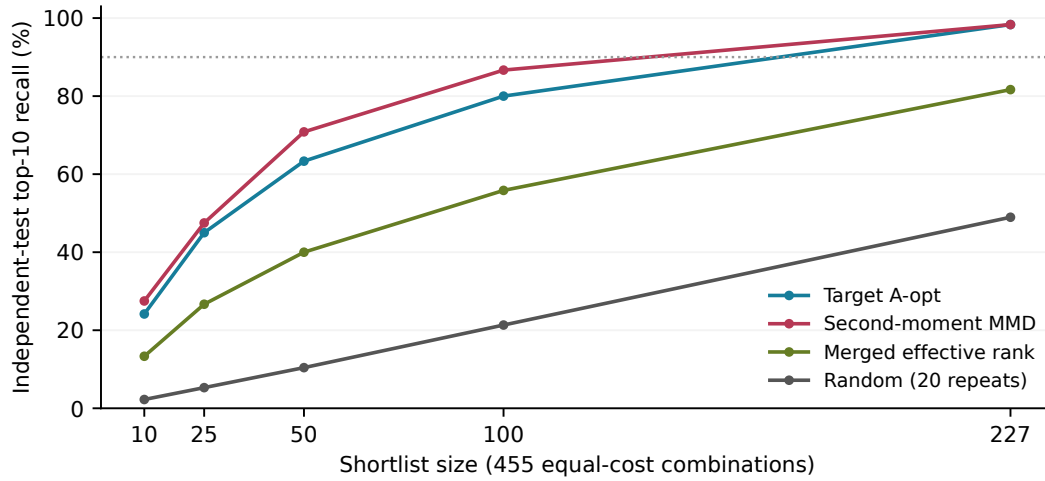


Figure 1: Mean test Brier top-ten recall is shown across shortlist sizes in the original DomainNet study. The dotted line marks the prespecified recall threshold. All curves use the same candidate combinations and feature access appropriate to each rule.

MMD reaches the same primary recall and selected regret as A-opt, and validation chooses the same final combination for both in every encoder–target task. At smaller shortlist sizes, MMD has higher mean recall (Figure 1). A-opt’s stronger full-ranking correlation therefore does not translate into better retention at those operating points. The common advantage over the tested target-blind rules points to the usefulness of target-conditioned second-order information; direct moment matching is already an effective way to use it here.

5.4 PROJECTION AND READOUT SENSITIVITY

The original ridge-softmax results have small differences between candidate losses. We therefore ask whether high recall persists when the projection changes and a different readout produces a wider range of losses. The exploratory follow-up uses four new projection seeds and logistic regression, keeping source membership and sample cost fixed. Table 2 reports the new-seed averages; the original seed is held aside as an implementation control.

A-opt retains 98.33% of the Brier top ten on average under each readout, with MMD slightly higher. The mean relative best-to-worst Brier span increases from 0.248% under ridge-softmax to 14.65% under logistic regression. High recall thus persists in a setting with more separation between candidates, rather than appearing only when their Brier losses are almost equal. The two target-aware scores also retain most NLL- and error-ranked candidates on average, with MMD higher in these comparisons.

Table 2: The exploratory DomainNet follow-up evaluates shortlists of size $s = 227$. Test top-ten recall (%) is averaged over two encoders and four new projection seeds within each target domain, then over six domains. Ridge denotes ridge-softmax. NLL and error use matching validation losses for final selection.

Readout	Metric	A-opt	MMD	Random
Ridge	Brier	98.33	99.38	50.12
Ridge	NLL	98.54	99.38	49.86
Ridge	Error	96.25	97.50	48.63
Logistic	Brier	98.33	98.96	48.97
Logistic	NLL	98.54	99.79	48.61
Logistic	Error	97.29	97.92	49.04

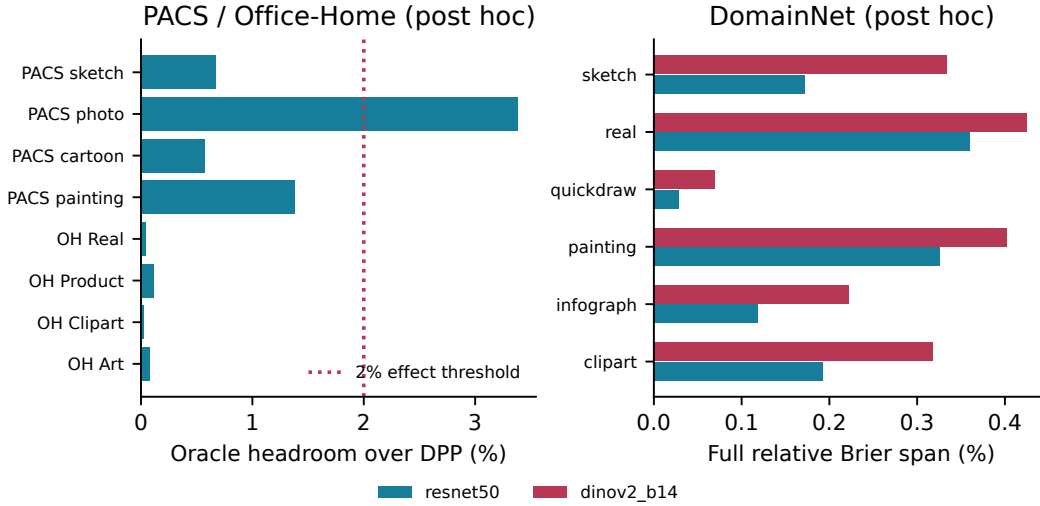


Figure 2: The panels report post-hoc diagnostics for different candidate families. The left panel shows oracle Brier improvement over DPP, averaging encoders within each PACS/Office-Home target. The right panel shows relative best-to-worst Brier spans for the original DomainNet tasks. The original evaluation thresholds are unchanged.

For Brier, both methods retain a test oracle in every new-seed task and have the same mean selected regret. Average retention does not describe every case: A-opt’s classification-error recall falls to 40% in one DINOv2–Quickdraw logistic task. Its shortlist still contains a test-error oracle, but validation chooses a model with error 1.66 percentage points higher. This example shows why broad retention, metric-specific retention, and final selection are reported separately. Appendix C gives the full comparator table and task-level details.

5.5 LOSS RESOLUTION AND TEST-SAMPLE STABILITY

The post-hoc loss-span diagnostic in Figure 2 puts the original regret criterion in context. Every candidate’s relative Brier regret lies below the 5% threshold, so that threshold cannot distinguish successful screens in this experiment. Recall provides the more informative comparison, but it raises another question: how sensitive is the test top ten to the sampled test examples?

A CUDA reconstruction first reproduces all original Brier top-ten sets. We then apply paired test-example bootstrap resampling while fixing fitted models and validation choices. Although resampled and original top-ten sets overlap by only 85.15% on average, A-opt and MMD retain 97.42% and 97.85% of the resampled top tens. Broad shortlists are less sensitive than exact rank membership to these test-sample changes.

Changing the metric has a larger effect on which combinations count as good: the original Brier and classification-error top tens overlap by only 29.17% on average. A screen should therefore be evaluated against the intended deployment metric, even when its average recall is high for several metrics. The resampling protocol and additional metric comparisons appear in Appendix B.

6 DISCUSSION

6.1 WHAT MAKES SCREENING USEFUL?

The results suggest evaluating screening in terms of the decision it enables. When the candidate family offers little improvement headroom, a small selected loss difference says little about the quality of a ranking. When candidates differ more, retaining several promising alternatives gives validation room to make a task-specific choice. A-opt and MMD both do this well on the audited DomainNet construction; their similar results make moment matching an important baseline for more elaborate geometric scores.

The natural operating point depends on the target metric, the acceptable loss gap, and the cost of supervised evaluation. Shortlist size controls how many candidates proceed, but total savings also depend on scoring, feature preparation, and validation costs. Separately, the sketch experiments in Appendix C show that valid score intervals can remain too wide to resolve choices at a useful byte budget. For a compressed score to support screening, it must still distinguish candidates at the intended shortlist budget.

6.2 LIMITATIONS

The equal-sample-cost evidence comes from one DomainNet construction. Its target domains share source pools, and the added seeds and readouts reuse the same images. They measure sensitivity within this benchmark rather than replication on independent data. Changes from PACS/Office-Home also include the dataset, projection, sample budget, and candidate construction, so they do not isolate candidate heterogeneity as a cause.

The Bayesian derivation concerns shared linear regression; the classification results establish the usefulness of its score empirically. Test-example resampling conditions on fitted models and validation choices. It does not measure uncertainty from training or validation selection. Absolute task quality also depends on the candidate family, as the supplementary adapter study illustrates. Finally, we have not measured matched end-to-end savings or ruled out overlap with encoder pre-training data. Independent equal-cost datasets, controlled validation budgets, and matched timing comparisons would address these remaining questions.

7 CONCLUSION

We developed a two-stage protocol that uses feature geometry to shortlist source-data combinations before supervised validation. The analysis explains how target weighting connects source coverage to risk under a shared linear model and separates screening omissions from final selection errors. Experiments show high candidate retention on equal-cost DomainNet combinations across projection seeds and readouts, with second-moment matching as effective as A-optimal screening at the primary operating point. Direct-selection results on PACS and Office-Home show that available improvement headroom also matters. Together, these findings support using target-aware geometry to narrow a search whose final decision is made with task-specific evidence. Measuring that decision's quality alongside the full evaluation cost is the next step toward choosing useful screening budgets.

AI USE STATEMENT

Generative AI tools assisted conceptual reformulation, mathematical claim and proof drafting, experiment design, software implementation, interpretation of results, literature search, manuscript writing, figure scripting, and Chinese translation. Numerical findings in this draft are taken from recorded experiment artifacts rather than generated observations. AI assistance is not independent validation of those artifacts or of the mathematical claims. Final verification and responsibility for a submitted version remain with the human authors.

REPRODUCIBILITY STATEMENT

Proof details appear in Appendix A; preprocessing, access rules, candidate construction, metrics, and result paths appear in Section 4 and Appendix B. The code repository is <https://anonymous.4open.science/r/collapsemodel-1DB6>. Outcome tables and manifests support aggregate reproduction; large feature caches are identified by hashes rather than duplicated in Git. Exact hardware-independent bitwise feature reproduction is not claimed.

ETHICS STATEMENT

The study uses existing benchmark images and pretrained encoders, with no new participant recruitment. Public availability does not remove image licensing, privacy, or representational-bias concerns. We do not claim that a diverse feature spectrum or a high shortlist recall makes a selected pool socially representative, fair, or private. The candidate taxonomy, pretraining overlap uncertainty, and restriction to visual classification limit downstream deployment claims. Dataset and model access remain subject to their original terms.

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A PROOFS AND SCOPE OF THE MATHEMATICAL CLAIMS

Let u, v be unit directions with $u^\top v = c \in [0, 1]$. In the orthonormal basis u, z , where $v = cu + \sqrt{1 - c^2}z$, the operator $a^2 uu^\top + b^2 vv^\top$ is

$$\begin{pmatrix} a^2 + b^2 c^2 & b^2 c \sqrt{1 - c^2} \\ b^2 c \sqrt{1 - c^2} & b^2 (1 - c^2) \end{pmatrix}. \quad (8)$$

Its trace is $a^2 + b^2$ and determinant $a^2 b^2 (1 - c^2)$, so its eigenvalues are

$$\lambda_{\pm} = \frac{a^2 + b^2 \pm \sqrt{(a^2 - b^2)^2 + 4a^2 b^2 c^2}}{2}. \quad (9)$$

At $c = 1$, the second branch is zero. These pairs give a full spectrum when the associated blocks are mutually orthogonal and invariant for both source Grams; unpaired directions contribute their own eigenvalues. This compatibility condition matters because principal vectors need not be singular vectors.

For fixed H_S and Gaussian likelihood, completing the square in the prior times likelihood gives posterior precision $P_0^{-1} + \sigma^{-2} H_S^\top H_S$ and mean $\mu_S = P_S \sigma^{-2} H_S^\top y_S$. For an independent target observation $y = h^\top \theta + \epsilon$, conditional on the training data and h ,

$$\mathbb{E}[(y - h^\top \mu_S)^2 \mid H_S, y_S, h] = \sigma^2 + h^\top P_S h. \quad (10)$$

Integrating h gives Equation (3). We assume the target covariate distribution does not reveal additional information about θ beyond this model. If a source has a different regression function, or labels are multiclass with softmax readout, this derivation no longer identifies the evaluated loss. The proof does not bound this misspecification error. Replacing C_T by a finite-sample estimate adds another discrepancy not removed by the identity.

If $G \succeq \tilde{G} \succeq 0$, inverse order implies $(P_0^{-1} + \sigma^{-2} G)^{-1} \preceq (P_0^{-1} + \sigma^{-2} \tilde{G})^{-1}$. Taking a trace against a PSD target moment preserves this order. More generally, suppose the sum of omitted sketch tails satisfies $0 \preceq G - \tilde{G} \preceq \delta I$. Define $B = P_0^{-1} + \sigma^{-2} \tilde{G}$. Then

$$\text{tr}(\hat{C}_T (B + \sigma^{-2} \delta I)^{-1}) \leq f_A(G) \leq \text{tr}(\hat{C}_T B^{-1}). \quad (11)$$

One admissible δ is the sum of valid spectral-norm bounds on each PSD tail. A candidate's proxy ordering is certified when its upper bound is below every competing lower bound. Thus interval width, as well as coverage, determines whether a sketch can resolve the next choice. The archived sweep evaluates this type of proxy comparison with its documented byte accounting.

Let S_Q minimize ℓ over $\mathcal{Q}_s$. Uniform deviation on that shortlist gives

$$\ell(\hat{S}) \leq v(\hat{S}) + \varepsilon \leq v(S_Q) + \varepsilon \leq \ell(S_Q) + 2\varepsilon. \quad (12)$$

Subtracting $\min_{S \in \mathcal{C}} \ell(S)$ proves Equation (6). No probability is asserted by this algebra. For population generalization, one must additionally justify the deviation event using the actual dependence and selection scheme. Exhaustive finite-test scores alone do not supply that justification.

The feature map $\phi(h) = hh^\top$, equipped with Frobenius inner product, satisfies $\langle \phi(h), \phi(h') \rangle_F = (h^\top h')^2$. The squared distance between source and target empirical feature means is therefore Equation (5). This is the biased empirical squared MMD (a V-statistic), not the diagonal-excluded unbiased estimator. It depends on uncentered second moments and cannot distinguish all distributions sharing those moments. We use it as a computationally simple same-information comparator, not a complete discrepancy measure.

B IMPLEMENTATION AND AUDIT DETAILS

The original DomainNet study fits a ridge readout and transforms its scores into class probabilities. Its weights and Brier loss are

$$\begin{aligned} W_S &= (I + G_S)^{-1} \sum_{i \in S} H_i^\top Y_i, & p(x) &= \text{softmax}(h(x)^\top W_S), \\ b(S) &= \frac{1}{n_T} \sum_t \|p(x_t) - e_{y_t}\|_2^2. \end{aligned} \quad (13)$$

Y_i is the one-hot source-label matrix over the fixed 100-class vocabulary. There is no intercept or temperature tuning. Linear algebra uses float64 after loading float32 features. For effective rank obtained through a Gram eigendecomposition, negative roundoff eigenvalues are clipped to zero, square roots form singular values, and only $\sigma_j > \sigma_{\max} \sqrt{8\epsilon_{64} \max(n_S, d, 1)}$ are retained. The zero matrix returns rank zero in code. This tolerance concerns numerical rank; the entropy definition itself is unchanged. The shared projection uses seed 20260911 and $d = 128$; no outcome-dependent dimension choice is used within the frozen DomainNet protocol.

Bayesian D-opt maximizes $\log \det(I + G_S)$. Subspace DPP takes each block’s top 16 Gram eigenvectors U_i , defines $K_{ij} = \|U_i^\top U_j\|_F^2 / 16$ with $K_{ii} = 1$, numerically projects the matrix to a PSD correlation matrix, and maximizes $\log \det(K_{S,S} + 10^{-8}I)$. Its kernel describes block subspaces, not individual-image features. Domain balance maximizes the number of represented domains minus 10^{-3} times the sum of squared domain block counts. Deterministic scores break ties lexicographically by combination identifier. These choices are fixed comparators; poor performance of this DPP construction is not a result about all determinantal selection methods.

DomainNet records image content hashes, sample identifiers, official split membership, role assignments, and class vocabulary. Archive and split checksums are validated before feature extraction. Disjoint role membership and duplicate-content checks address overlap within the materialized benchmark; they do not rule out semantic near-duplicates or pretraining contamination. All screen methods use the same source blocks, projection, target-selection rows, and candidate family. Feature extraction ignores labels, while benchmark materialization uses taxonomy metadata and stores labels for later authorized stages.

Screen, validation, and audit manifests bind configuration and upstream artifact hashes. Stage-specific loaders enforce label and split permissions. In the original protocol, validation evaluates each method or random repeat separately. The test audit covers all 6×455 combinations per encoder after choices are frozen; its exhaustive oracle serves as a research reference. The follow-up caching policy is described below.

The CUDA reconstruction retains the original sample identifiers, candidate membership, preprocessing, and hyperparameters. All twelve Brier top-ten sets agree with the original study. Per-example loss means reproduce the reconstructed aggregate audit within 1.78×10^{-15} . For each target domain, 2,000 paired bootstrap draws resample its 2,048 test examples, sharing weights across combinations, encoders, and four losses. Fitted models and validation choices stay fixed; combination identifiers resolve ties. These draws measure sensitivity to test-example sampling. The original Brier top ten overlaps with NLL by 90.83%, raw squared loss by 30.83%, and classification error by 29.17% on average.

Seeds 20260917, 20260918, 20260919, and 20260920 generate new 128-dimensional Rademacher projections with the same pre/post row normalization. Seed 20260911 is an implementation control, excluded from new-seed means. Logistic regression uses L2 regularization, $C = 1$, no intercept, LBFGS, tolerance 10^{-6} , and at most 1,000 iterations. A convergence warning aborts a task; no such warning occurred, and the largest iteration count was 24. The class vocabulary is fixed, with zero probability assigned to source-absent classes.

All methods retain their screening rules, candidate combinations, and five shortlist sizes. Separate processes screen, fit and validate, then test frozen models. The audit checks source, configuration, lineage, model, and selection hashes before reading test features. The follow-up fits every combination once per readout and reuses these fits across methods. This cached exhaustive evaluation supports counterfactual comparisons of shortlists. Brier, NLL, and error use their own validation losses for final selection. Original-seed ridge reproduces 5,460 reconstructed test combinations with maximum loss difference 1.78×10^{-15} and all 1,620 Brier validation choices exactly.

The two encoders and five seeds produce 54,600 fits across both readouts. New-seed summaries first average random repeats within a task, then encoders and seeds within each target domain, and finally the six domains. Each readout contributes 48 repeated encoder–seed–domain measurements. They remain six domain units sharing the same source pools.

The archive retains historical experiment IDs to identify results without confusing them with the paper’s scientific questions. The following paths are relative to the anonymous code repository:

- Shared-model and shift studies (E1): `results/target_conditioned/e1_synthetic/expanded_v3_controlled_rng/`.
- PACS/Office-Home comparison (E2): `results/target_conditioned/e2_data_set_selection/method_v1/`.
- Post-hoc headroom (E2a): `results/target_conditioned/e2_dataset_selection/oracle_headroom_v1/`.
- Equal-cost shortlist (E2b): `results/target_conditioned/e2b_fixed_cost_shortlist/domainnet_v1/`.
- Risk sketch sweep (E5): `results/target_conditioned/e5_sketch_certificate/expanded_v3_controlled_rng/`.
- Reconstruction and test resampling (R1/R2): `results/target_conditioned/revision_20260916/`.
- Projection and readout follow-up (R3/R4): `results/target_conditioned/revision_20260917/`.

The DomainNet data manifest begins `1ba25059df70be4d`, candidate-set ID `545fe86c8c9e8304`, and summary ID `2d473cb9eb977e85`; full SHA-256 identifiers are in the artifacts. The original configuration is `code/configs/target_conditioned_e2b/domainnet_v1.json`, implemented in `scripts/run_target_conditioned_e2b_shortlist.py`. The follow-up uses `code/configs/target_conditioned_e2b/readout_projection_v1.json` and `scripts/run_e2b_readout_projection.py`. Its summary tables are in `revision_results/e2b_readout_projection_v1/summary/` under the R3/R4 directory. The manuscript’s `build_assets.py` regenerates tables, figures, numeric macros, and bilingual abstracts from recorded sources, logging input hashes in `generated/asset_manifest.json`.

The three original feature matrices are not duplicated in Git. Metadata record their exact file and checkpoint hashes, shapes, preprocessing, extraction code revision, and runtime. Re-extraction requires access to the original public data and model weights; bitwise identity across different hardware is not assumed. The repository contains manifests and sufficient outcome tables to reproduce reported aggregate results without those matrices.

C SUPPLEMENTARY RESULTS

At $K = 3$ on PACS/Office-Home, mean Brier is 0.826769 for A-opt, 0.829162 for DPP, and 0.828719 for MMD. The paired A-opt-minus-baseline intervals are $[-0.007449, 0.000731]$ and $[-0.003392, -0.000712]$, respectively. The post-hoc audit covers all 220 combinations per task. Oracle headroom over DPP has descriptive 95% interval $[0.162, 1.617]\%$; only PACS photo exceeds the original 2% threshold at the target-unit level. The oracle improves on the post-hoc test-best envelope of parent methods by 0.087% on average. This envelope is an outcome-based reference, not a selectable baseline.

Table 3 gives the complete Brier recall comparison at the primary size. A-opt and MMD have the same mean relative selected regret: 0.001006% for ridge and 0.103% for logistic regression. All their new-seed primary Brier shortlists contain an empirical oracle.

The lowest A-opt classification-error recall occurs for DINOv2, Quickdraw, logistic regression, and projection seed 20260918. The test-error oracle is 85.9863%, compared with 87.6465% for the validation choice. Its oracle remains in the shortlist despite the 40% top-ten recall. Thus retaining the best member and retaining the best ten answer different questions, and both are distinct from the absolute quality of the available models.

The four-encoder, ten-construction-seed spectral sweep uses six overlap settings and four budgets, with 960 configurations per sketch rank. At ranks 10, 20, 50, relative exact merged-effective-rank gains over rank-only are respectively 2.09, 3.81, 4.94% for the Gram sketch and 4.90, 5.48, 5.68% for DPP. These gains are in the geometry objective, not classification accuracy. Its entropy-continuity bounds follow the type of bound in Audenaert (2007); all 2,880 selected-prefix intervals cover, but

Table 3: Brier top-ten recall (%) is reported for all follow-up scoring rules at $s = 227$. Averages use the same domain-level procedure as Table 2; original-seed controls are excluded.

Screen	Ridge	Logistic
Target A-opt	98.33	98.33
Second-moment MMD	99.38	98.96
Bayesian D-opt	82.29	86.46
Merged effective rank	80.42	85.42
Random (20 repeats)	50.12	48.97
Target energy	41.04	33.96
DPP subspace	28.75	35.42
Domain balance	26.46	31.67

none of 7,200 evaluated greedy steps is certified. The archived source is `results/two_stage_controlled_rank_sweep_corrected_v3/`. These observations do not imply that every low-rank sketch must fail.

The separate E5 study compares sketch payloads to packed symmetric float64 Grams, not inflated dense storage. Among full-grid strategies whose mean byte ratio is at most 25%, the best selected-set match to Full-Gram greedy is 15.42%, achieved by adaptive rank cap 2 with mean byte ratio 14.91%. Its pairwise certificate rate is 3.91% and fully certified selection rate zero. Adaptive fallback reaches full selected-set agreement but costs approximately 160% of packed-Gram bytes on average when previous transmissions are counted. This fails the joint decision-and-communication requirement. The reference is Full-Gram greedy, not the exhaustive global minimizer.

An earlier, different experiment trains source-supervised bottleneck adapters on 21 sources and audits seven targets with four encoders and five adapter seeds. Target-train validation uses three stratified five-fold partitions, with held-out target test accessed separately after selection freezing. Frozen identity reaches mean accuracy 0.83654, an untrained random adapter 0.77144, and the validation-oracle source-supervised adapter 0.76951. The last trails identity by 0.06703, with target-cluster 95% interval $[-0.09771, -0.04400]$. Small shortlist regret relative to this adapter family is therefore not downstream benefit. This is not a DomainNet experiment or evidence against all adaptation methods. Results reside in `results/two_stage_repeated_cv_utility_corrected_v3/`.